

Visualizing Data

Part of **Data Science Starter** · Estimated time: 35 min

Introduction

A well-chosen chart can show in one glance what would take paragraphs to describe in words. The trick is matching the chart type to the kind of comparison you're actually making.

Core ideas

- Use a bar chart to compare a handful of categories, a line chart to show change over time, and a scatter plot to look for a relationship between two numeric variables.
- Every chart should be labeled: axis titles with units, a legend if there's more than one series, and a clear title stating what the chart shows.
- Sorting bars by value (instead of alphabetically) usually makes a comparison easier to read at a glance.
- More color or decoration doesn't make a chart more informative — it often hides the pattern you're trying to show.

Example

```
import matplotlib.pyplot as plt
df.groupby("region")["revenue"].sum().plot(kind="bar")
plt.ylabel("Revenue ($)")
plt.title("Revenue by Region")
plt.show()
```

Try it yourself

Using the dataset from the pandas module, make one bar chart and one line chart (if you have any date-like column) with proper axis labels and a title.

Further reading

- [Matplotlib — Tutorials](https://matplotlib.org/stable/tutorials/index.html)

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- [From Data to Viz — chart chooser](https://www.data-to-viz.com/)

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